

High-Energy Task Library Development: Approved Task Libraries

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The **Edison Electric Institute** (EEI) is the association that represents all U.S. investor-owned electric companies. Our members provide safe, reliable electricity for nearly 250 million Americans, and operate in all 50 states and the District of Columbia. Collectively, the electric power industry supports more than 7 million jobs in communities across the United States and drives economic growth and prosperity. EEI also includes hundreds of industry suppliers and related organizations as Associate Members. Safety is a core value of the electric power sector and EEI members are committed to protecting the safety and health of the workforce, industry contractors, and customers. EEI members have prioritized SIF elimination and through collaboration across the industry are implementing a series of actions to achieve this goal through the [Power to Prevent SIF Initiative](#).

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EEI Task Library Working Group Overview

EEI Working Group for Standardized Task Libraries

Objective: *Align on work breakdown structure and develop a standardized list of tasks for all Electric and Gas Utility work to enable operationalizing and benchmarking of Safety Data*

Contributing Organizations:

Alliant Energy	Eversource Energy
Ameren	Exelon Corp
American Gas Association (AGA)	Itron
American Transmission Co	National Grid
Arizona Public Service Co	NiSource
Bruce Power	Public Service Electric & Gas
CenterPoint Energy	Southern California Edison
Construction Safety Research Alliance (CSRA)	Utility Vegetation Services LLC
Consumers Energy	
Dominion Energy	
Duke Energy	
Emera	
Entergy Corp	

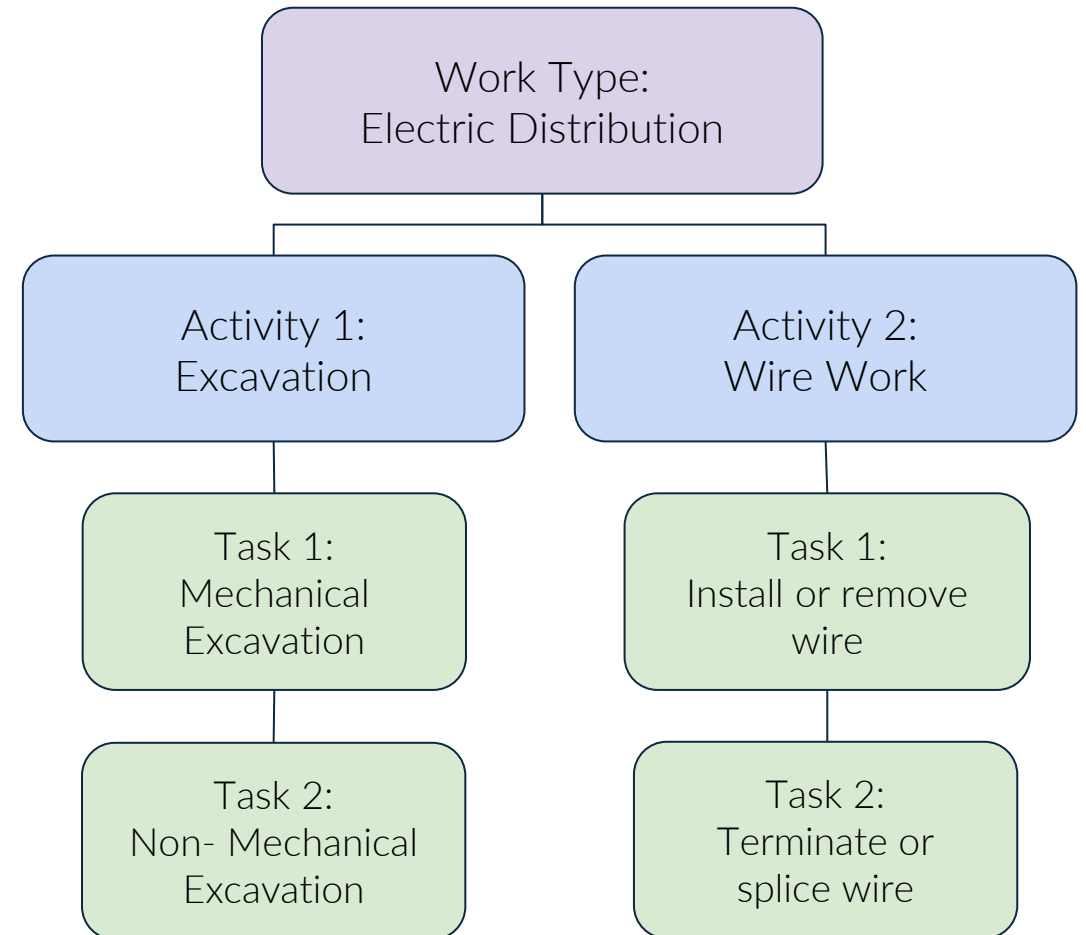
Scope of Work:

- *Electric Distribution, Transmission, Substation*
- *Gas Plant & Station, Gas Distribution & Regulator Station*
- *Electric Generation: Gas, Nuclear, Diesel, Solid Fuel*
- *Vegetation Management, Fleet Work*
- *Supply Chain and Warehouse*
- *Renewable Generation: Solar, Wind, Hydro, Battery*

Activity and Task Definitions

What is an Activity?

- An “Activity” is a grouping of tasks which can be customized for different use cases. The purpose of “activity” in the task library hierarchy is to improve the user experience by grouping tasks in ways they find most useful.
- Example Activities: *Excavation*, *Pole work*, *Cable work*, *Wire work*, *Operate equipment*
- Activities are typically organized around the following:
 - Types of equipment (e.g. pole work, wire work)
 - Types of work (e.g. prep work site, material handling)



What is a Task?

A task is a combination of the action performed (e.g. install, remove, etc.), the asset that the action is performed on (material), and the work method used (tools and competency)

Task → Action Performed + Asset + Work Method (if needed) → E.g. Install or remove cable brackets

Combine into a single task if:

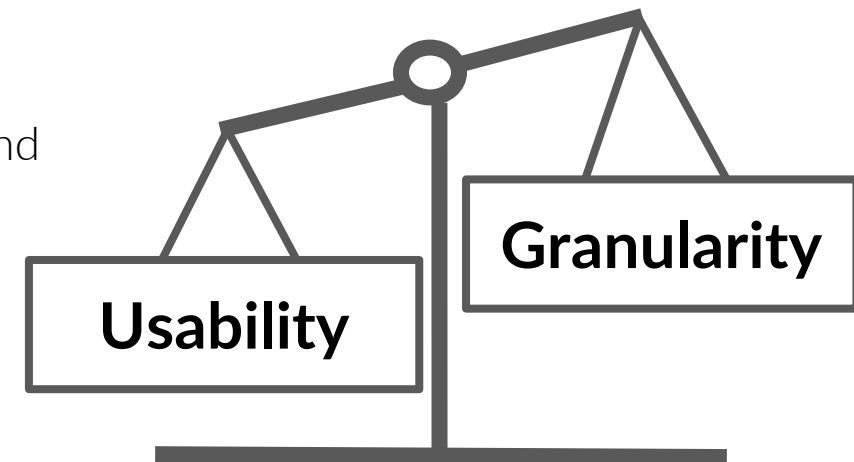
- Different actions on the same asset have the same hazards
- Change in type of asset (material) does not impact hazards and controls

Separate into different tasks if:

- Task is the same (excavating) but can have multiple methods (tools and competency) with different hazards and controls

Other rules for definition and granularity:

- Some tasks are asset agnostic and limited to the action and work method (e.g. phasing, critical lift, etc.)
- At times, common industry verbiage is used for naming tasks to enhance usability (e.g. confined space entry)



*Task definition requires
balancing granularity with
usability*

Example Task - Combined Action

TASK → Action + Asset + Work Method

Action

Install AND remove work area protection (i.e. traffic control)

- Install and remove on the same asset
- No change to hazards based on actions

Install work area protection



Remove work area protection



Two tasks combined due to similar tools, competency, and hazards

Example Task - Combined Assets

TASK → Action + Asset + Work Method

Asset

Install or remove equipment or devices (i.e. transformer, sectionalizer, recloser, cutout, switch, arrestor, insulator, regulator, capacitor) - Overhead

- Change in asset/material/equipment
- No change to hazards based material or equipment

Install transformer



Install switch



Two tasks combined due to similar tools, competency, and hazards

Example Task - Separate Work Methods

TASK → Action + Asset + Work Method

Work Method

- Mechanical excavation (e.g. excavator, digger derrick)
- Non-mechanical excavation (e.g. vac truck, hand dig)

- Action is the same (excavating)
- Different methods require different tools/competency

Mechanical excavation



Non-mechanical excavation



Two tasks are separate due to different tools, competency, and hazards

Electric Distribution

Electric Distribution Task Library (1 of 4)

Activity	Task
Excavate	Backfill (tamping, compacting, soil dumping)
	Grade (e.g. skid steer, bobcat, etc.)
	Install or remove protective systems (e.g. shoring, shielding, benching, sloping)
	Install or remove tieback
	Mechanical excavation (e.g. excavator, digger derrick, saw cutting, etc.)
	Non-mechanical excavation (e.g. vacuum truck, hand dig, etc.)
	Restore soft surface (e.g. seed, mulch, lime, sod, etc.)
	Sample soil
Prep work site	Assemble or demo scaffold
	Clear and remove debris, vegetation, trees, or obstructions
	Install or remove fall protection system
	Install or remove live line insulation, cover up, or rubber goods
	Install or remove matting
	Install or remove work area protection (i.e. traffic control)
	Locate or mark-out
	Operate work area protection (i.e. flaggers)
	Survey and stake
	Install or remove arc flash blankets
	Load or unload material or equipment

Electric Distribution Task Library (2 of 4)

Activity	Task
Material handling	Load or unload material or equipment
	Rig and lift - Non-critical lift
	Rig and lift - Critical lift with plan in place
Manhole or vault work	Enter or work in confined or enclosed space
	Install or remove cable brackets
	Install or remove vault, manhole, or pad
	Maintain or inspect vault, manhole, or pad
	Repair vault, manhole, or pad

Electric Distribution Task Library (3 of 4)

Activity	Task
Pole work	Climb pole
	Inspect pole
	Install or remove anchor or guy
	Install or remove live line insulation, cover up, or rubber goods
	Install or remove pole, crossarm, or brackets
Cable work	Fault locate on cable (e.g. HIPOT)
	Install or remove arc flash blankets
	Install or remove cable brackets
	Install or remove conduit or boring
	Install, pull, maintain, or remove cable or elbow
	Terminate and splice cable
	Test, maintain, inspect, or troubleshoot cable
Wire work	Fault locate on wire
	Install or remove live line insulation, cover up, or rubber goods
	Install, remove, or transfer wire
	Terminate and splice wire
	Test, maintain, inspect, patrol, or troubleshoot wire
	Install or remove grounding
Meter work	Install or remove current transformer (CT) and potential transformer (PT) metering
	Install or remove live line insulation, cover up, or rubber goods
	Install or remove meter
	Test or inspect meter

Electric Distribution Task Library (4 of 4)

Activity	Task
Work on electrical equipment and devices	Install or remove equipment or devices (i.e. transformer, sectionalizer, recloser, capacitor) - Underground
	Install or remove equipment or devices (i.e. transformer, sectionalizer, recloser, cutout, switch, arrestor, insulator, regulator, capacitor) - Overhead
	Install or remove fault indicator
	Install or remove live line insulation, cover up, or rubber goods
	Install, remove, or repair pad or foundation
	Test, maintain, inspect, or troubleshoot equipment or devices (sectionalizer, recloser, switch, arrestor, insulator, regulator, capacitor, or distribution xfmr)
	Install or remove grounding
Operate equipment	Change settings / taps on device (e.g. regulator, inverter, capacitor, etc.)
	Conduct load check or phasing
	Connect or pull elbows
	Cut conductors in the clear
	Cut in floats or bells
	Energize or de-energize a circuit
	Install or remove grounding
	Install or remove live line insulation, cover up, or rubber goods
	Install or remove taps
	Open, close, or bypass cutout, switch, recloser, fuse, or sectionalizer

Electric Transmission

Electric Transmission Task Library (1 of 4)

Activity	Task
Excavate	Backfill (tamping, compacting, soil dumping)
	Grade (e.g. skid steer, bobcat, etc.)
	Install or remove protective systems (e.g. shoring, shielding, benching, sloping)
	Install or remove tieback
	Mechanical excavation (e.g. excavator, digger derrick, saw cutting, etc.)
	Non-mechanical excavation (e.g. vacuum truck, hand dig, etc.)
	Restore soft surface (e.g. seed, mulch, lime, sod, etc.)
Prep work site	Sample soil
	Assemble or demo scaffold
	Clear and remove debris, vegetation, trees, or obstructions
	Install or remove fall protection system
	Install or remove live line insulation, cover up, or rubber goods
	Install or remove matting
	Install or remove work area protection (i.e. traffic control)
	Operate work area protection (i.e. flaggers)
	Locate or mark-out
	Survey and stake
	Install or remove arc flash blankets
	Load or unload material or equipment



Electric Transmission Task Library (2 of 4)

Activity	Task
Material handling	Load or unload material or equipment
	Rig and lift - Non-critical lift
	Rig and lift - Critical lift with plan in place
Manhole or vault work	Enter or work in confined or enclosed space
	Install or remove cable brackets
	Install or remove vault, manhole, or pad
	Maintain or inspect vault, manhole, or pad
	Repair vault, manhole, or pad

Electric Transmission Task Library (3 of 4)

Activity	Task
Wire work	Install, sag, dead end, transfer, or remove wire
	Terminate and splice wire
	Test, maintain, inspect, or troubleshoot wire
	Fault locate on wire
	Aerial work from helicopter
	Aerial inspection from helicopter
	Install or remove live line insulation, cover up, or rubber goods
	Install or remove grounding
Cable work	Install or remove conduit, duct bank, or cable trenches
	Install, pull, maintain, or remove cable or elbow
	Terminate and splice cable
	Install or remove cable brackets
	Test, maintain, inspect, or troubleshoot cable
	Fault locate on cable (e.g. HIPOT)
	Install or remove arc flash blankets



Electric Transmission Task Library (4 of 4)

Activity	Task
Structure work	Install or remove transmission structure (e.g. tower, pole)
	Install or remove down guys
	Install or remove cathodic protection system
	Install, remove, or repair pad or foundation
	Inspect and maintain structure and foundation
	Install, remove, or maintain tower lights
	Climb structure
	Aerial work from helicopter
	Aerial inspection from helicopter
	Install or remove live line insulation, cover up, or rubber goods
	Install or remove fall protection system
	Install, remove, or replace fencing
	Install or remove grounding
Work on line hardware or devices	Install or remove arrestor
	Install or remove capacitor or reactor
	Install or remove insulator
	Install or remove switch
	Install or remove cathodic protection system
	Install or remove aerial markers
	Test, maintain, inspect, or troubleshoot arrestor
	Test, maintain, inspect, or troubleshoot capacitor or reactor
	Test, maintain, inspect, or troubleshoot insulator
	Test, maintain, inspect, or troubleshoot switch
	Test, maintain, inspect, or troubleshoot cathodic protection system
	Install or remove grounding



Electric Substation

Electric Substation Task Library (1 of 5)

Activity	Task
Excavate	Backfill (tamping, compacting, soil dumping)
	Grade (e.g. skid steer, bobcat, etc.)
	Install or remove protective systems (e.g. shoring, shielding, benching, sloping)
	Install or remove tieback
	Mechanical excavation (e.g. excavator, digger derrick, saw cutting, etc.)
	Non-mechanical excavation (e.g. vacuum truck, hand dig, etc.)
	Restore soft surface (e.g. seed, mulch, lime, sod, etc.)
	Sample soil
Prep work site	Assemble or demo scaffold
	Clear and remove debris, vegetation, trees, or obstructions
	Install or remove fall protection system
	Install or remove live line insulation, cover up, or rubber goods
	Install or remove matting
	Install or remove work area protection (i.e. traffic control)
	Operate work area protection (i.e. flaggers)
	Locate or mark-out
	Survey and stake
	Install or remove arc flash blankets
	Load or unload material or equipment



Electric Substation Task Library (2 of 5)

Activity	Task
Material handling	Load or unload material or equipment
	Rig and lift - Non-critical lift
	Rig and lift - Critical lift with plan in place
Manhole or vault work	Enter or work in confined or enclosed space
	Install or remove cable brackets
	Install or remove vault, manhole, or pad
	Maintain or inspect vault, manhole, or pad
	Repair vault, manhole, or pad

Electric Substation Task Library (3 of 5)

Activity	Task
Foundation and civil work	Install, remove, or repair pad or foundation
	Install, remove, or repair spill containment
	Install or remove conduit, duct bank, or cable trenches
	Fabricate, alter, assemble, or remove structure
	Install, remove, or repair fencing
Ground grid work	Install, remove, or repair ground grid
	Test ground grid
Metal clad switchgear work	Install or remove metal clad switchgear
	Test, maintain, or inspect bus work (metal clad switchgear)
	Test, maintain, or inspect circuit breaker (metal clad switchgear)
	Test, maintain, or inspect PT and CT (metal clad switchgear)
	Test, maintain, or inspect switchgear cubicle (metal clad switchgear)
	Test, maintain, or inspect ground and test device (metal clad switchgear)
Wire and bus work	Install or repair equipment grounding lead
	Install, remove, or maintain bus work (flexible / hard)
	Install or remove primary wire
	Install or remove grounding
Cable work	Install or remove conduit, duct bank, or cable trenches
	Install, pull, maintain, or remove cable or elbow
	Terminate and splice cable
	Install or remove cable brackets
	Test, maintain, inspect, or troubleshoot cable
	Fault locate on cable (e.g. HIPOT)
	Install or remove arc flash blankets



Electric Substation Task Library (4 of 5)

Activity	Task
Work on equipment and devices (open air)	Install or remove transformer or regulator
	Install or remove circuit breaker
	Install or remove capacitor bank
	Install or remove switch, ground switch, or motor operated disconnect (MOD)
	Install or remove wiring, CCVT, CT, PT, or lightning arrestor
	Install or remove wave trap
	Install or remove reactor
	Install, remove, or repair cable trays
	Test, maintain, inspect, or troubleshoot transformer or regulator
	Test, maintain, inspect, or troubleshoot circuit breaker
	Test, maintain, inspect, or troubleshoot capacitor bank
	Test, maintain, inspect, or troubleshoot switch, ground switch, or motor operated disconnect (MOD)
	Test, maintain, inspect, or troubleshoot wiring, CCVT, CT, PT, or lightning arrestor
	Test, maintain, inspect, or troubleshoot wave trap
	Test, maintain, inspect, or troubleshoot reactor
	Install or remove grounding
Relay work	Install or remove relay
	Install or remove relay panel
	Install or remove station batteries and chargers
	Test, maintain, inspect, or troubleshoot relay
	Test, maintain, inspect, or troubleshoot relay panel
	Test, maintain, inspect, or troubleshoot station batteries and chargers



Electric Substation Task Library (5 of 5)

Activity	Task
Operate equipment	Conduct load check-or phasing
	Install or remove grounding
	Open or close circuit breaker
	Open or close switch, ground switch, or motor operated disconnect (MOD)
	Change taps on device (e.g. transformer, load tap changer, regulator, etc.)
	Rack-in or rack-out circuit breaker



Gas Distribution & Regulator Station

Gas Distribution and Reg. Station Task Library (1 of 4)

Activity	Task
Excavate	Backfill (tamping, compacting, soil dumping)
	Grade (e.g. skid steer, bobcat, etc.)
	Install or remove protective systems (e.g. shoring, shielding, benching, sloping)
	Install or remove tieback
	Mechanical excavation (e.g. excavator, digger derrick, etc.)
	Non-mechanical excavation (e.g. vacuum truck, hand dig, etc.)
	Restore soft surface (e.g. seed, mulch, lime, sod, etc.)
	Install, remove, or repair paving or asphalt
	Sample soil
Prep work site	Assemble or demo scaffold
	Clear and remove debris, vegetation, trees, or obstructions
	Install or remove fall protection system
	Install or remove matting
	Install or remove work area protection (i.e. traffic control)
	Operate work area protection (i.e. flaggers)
	Locate or mark-out
	Survey and stake
	Load or unload material or equipment
	Install or remove road plates



Gas Distribution and Reg. Station Task Library (2 of 4)

Activity	Task
Material handling	Load or unload material or equipment
	Rig and lift - Non-critical lift
	Rig and lift - Critical lift with plan in place
Manhole or vault work	Enter or work in confined or enclosed space
	Install or remove vault, manhole, or pad
	Maintain or inspect vault, manhole, or pad
	Repair vault, manhole, or pad

Gas Distribution and Reg. Station Task Library (3 of 4)

Activity	Task
Piping / coolant piping and valve work	Install or remove pipe, flange, T, bypass, or valve - Welding
	Install or remove plastic pipe, flange, bypass, T, or valve - Electric fusion
	Install pipe - Boring or horizontal directional drilling (HDD)
	Install pipe - Pneumatic piercing tools (e.g. missile or gopher)
	Install pipe - Lowering in
	Coat pipe or welds
	Test pipe welds - Non-destructive examination (NDE)
	Test, maintain, inspect, or troubleshoot pipe, flange, T, bypass, or valve
	Pressure test pipe (hydrostatic / pneumatic)
	Pigging pipe
	Field bend pipe
	Repair or replace pipe, flange, T, bypass, or valve
	Leak survey or investigation
	Install or remove cathodic protection
	Test, maintain, inspect, or troubleshoot cathodic protection system
	Tie-in or hot-tap pipe
Regulator station work	Install, remove, or repair station regulator
	Test, maintain, inspect, or troubleshoot station regulator
	Install or remove pump or compressor
	Test, maintain, inspect, or troubleshoot pump



Gas Distribution and Reg. Station Task Library (4 of 4)

Activity	Task
Customer meter work	Relight, test meter, check pilot, check draft, or check leak with CGI detector
	Install, remove, or repair customer meter
Electrical equipment work	Install or remove cable tray and support
	Test electrical equipment
	Install or remove wiring
	Install or remove conduit
Structure and civil work	Install, remove, or repair pad or foundation (concrete work)
	Install, remove, or repair spill containment
	Fabricate, alter, assemble, or remove structure
	Install, remove, or repair paving or asphalt
	Install, remove, or replace fencing
Operate gas infrastructure	Gas-in or gas-out
	Purge gas
	Open or close valves
	Operate regulator
	Operate bypass on pipe
	Pump drip pots



Gas Plant and Gate Station

Gas Plant and Gate Station Task Library (1 of 5)

Activity	Task
Excavate	Backfill (tamping, compacting, soil dumping)
	Grade (e.g. skid steer, bobcat, etc.)
	Install or remove protective systems (e.g. shoring, shielding, benching, sloping)
	Install or remove tieback
	Mechanical excavation (e.g. excavator, digger derrick, etc.)
	Non-mechanical excavation (e.g. vacuum truck, hand dig, etc.)
	Restore soft surface (e.g. seed, mulch, lime, sod, etc.)
	Install, remove, or repair paving or asphalt
	Sample soil
Prep work site	Assemble or demo scaffold
	Clear and remove debris, vegetation, trees, or obstructions
	Install or remove fall protection system
	Install or remove matting
	Install or remove work area protection (i.e. traffic control)
	Operate work area protection (i.e. flaggers)
	Locate or mark-out
	Survey and stake
	Load or unload material or equipment
	Install or remove road plates



Gas Plant and Gate Station Task Library (2 of 5)

Activity	Task
Material handling	Load or unload material or equipment
	Rig and lift - Non-critical lift
	Rig and lift - Critical lift with plan in place
Manhole or vault work	Enter or work in confined or enclosed space
	Install or remove vault, manhole, or pad
	Maintain or inspect vault, manhole, or pad
	Repair vault, manhole, or pad

Gas Plant and Gate Station Task Library (3 of 5)

Activity	Task
Piping / coolant piping and valve work	Install or remove pipe, flange, T, bypass, or valve - Welding
	Install or remove plastic pipe, flange, T, bypass, or valve - Electric fusion
	Install pipe - Boring or horizontal directional drilling (HDD)
	Install pipe - Pneumatic piercing tools (e.g. missile or gopher)
	Install pipe - Lowering in
	Coat pipe or welds
	Test pipe welds - Non-destructive examination (NDE)
	Test, maintain, inspect, or troubleshoot pipe, flange, T, bypass, or valve
	Pressure test pipe (hydrostatic / pneumatic)
	Pigging pipe
	Field bend pipe
	Repair or replace pipe, flange, T, bypass, or valve
	Leak survey or investigation
	Install or remove cathodic protection
	Test, maintain, inspect, or troubleshoot cathodic protection system
	Tie-in or hot tap pipe
Odorizer work	Install or remove odorizer equipment
	Load or unload odorant
	Test, maintain, inspect, or troubleshoot odorizer



Gas Plant and Gate Station Task Library (4 of 5)

Activity	Task
Heat exchanger work	Install, remove, or repair heat exchanger
	Test, maintain, inspect, or troubleshoot heat exchanger
	Light heat exchanger
Station meter or regulator work	Install, remove, or repair station meter or regulator
	Test, maintain, inspect, or troubleshoot station meter or regulator
Pump or compressor work	Install, remove, or repair pump or compressor
	Maintain, inspect, or troubleshoot pump or compressor
Storage vessel work	Install, remove, or repair storage vessel
	Maintain, inspect, or troubleshoot storage vessel
Generator work	Install, remove, or repair generator
	Maintain, inspect, or troubleshoot generator
Fire suppression system work	Test, maintain, inspect, or troubleshoot fire suppression system
	Install, remove, repair, or replace fire suppression system
Electrical equipment work	Install or remove cable tray and support
	Test electrical equipment
	Install or remove wiring
	Install or remove conduit
Structure and civil work	Install, remove, or repair pad or foundation (concrete work)
	Install, remove, or repair spill containment
	Fabricate, alter, assemble, or remove structure
	Install, remove, or replace paving or asphalt
	Install, remove, or replace fencing



Gas Plant and Gate Station Task Library (5 of 5)

Activity	Task
Operate gas infrastructure	Gas-in or gas-out
	Purge gas
	Open or close valves
	Operate regulator
	Operate bypass on pipe
	Pump drip pots
	Operate pumps
	Operate vaporizers
	Operate heaters



Generation - Gas Simple Cycle

Generation - Gas Simple Cycle Task Library (1 of 6)

Activity	Task
Fuel system work	Inspect or maintain fuel storage tanks
	Inspect and calibrate fuel transfer system
	Calibrate fuel system instrumentation
	Repair or replace pipe, flange, T, bypass, or valve - Welding
	Coat pipe or welds
	Test pipe welds - Non-destructive examination
	Test, maintain, inspect, or troubleshoot pipe, flange, T, bypass, or valve
	Test, maintain, inspect, or troubleshoot cathodic protection system
	Pressure test pipe (hydrostatic / pneumatic)
	Leak survey, response, or investigation
	Unload fuel
	Maintain fuel pumps and valves
	Replace gaskets, filters, or regulating valve diaphragms
Water injection system work	Maintain water pumps and valves
	Replace gaskets, filters, or regulating valve diaphragms
	Calibrate water injection system instrumentation
	Operate water treatment system
	Test, maintain, inspect, or troubleshoot water injection system

Generation - Gas Simple Cycle Task Library (2 of 6)

Activity	Task
Continuous emission monitoring system (CEMS) work	Calibrate emissions sensors
	Replace CEMS test gases
	Test CEMS (i.e. stack test, RATA test)
	Repair or replace CEMS umbilical
	Test, maintain, inspect, or troubleshoot water cooling system
	Test, maintain, inspect, or troubleshoot CEMS analyzers
Intake air system work	Replace intake air filters
	Calibrate intake air instrumentation
	Test, maintain, inspect, or troubleshoot intake air system
	Test, maintain, inspect, or troubleshoot evaporative cooling system
	Test, maintain, inspect, or troubleshoot inlet bleed heat system
	Test, clean, or inspect inlet guide vanes
Backup power system work	Repair or replace battery system
	Repair or replace control system
	Test, maintain, service, inspect, or troubleshoot battery system
	Measure voltage and current on battery bank
	Test, maintain, inspect, or troubleshoot control system
	Install, remove, or replace control panel and communication system
	Test, maintain, or inspect emergency generator

Generation - Gas Simple Cycle Task Library (3 of 6)

Activity	Task
Turbine generator system - Turbine work	Repair or replace turbine
	Teardown and rebuild turbine
	Test, maintain, inspect, or troubleshoot turbine
	Test, maintain, inspect, or troubleshoot turbine rotor
	Test, maintain, inspect, or troubleshoot turbine fuel and lubrication oil system
	Test, maintain, inspect, or troubleshoot turbine compressor
	Perform offline water wash of turbine compressor
	Calibrate turbine instrumentation
	Balance turbine rotor
	Check turbine alignment
	Calibrate turbine equipment
	Replace gaskets and filters on fuel and lube oil lines
	Align turbine for operations, startup, and shutdown
	Repair or replace combustor, fuel nozzles, or exhaust system
	Test, maintain, inspect, or troubleshoot combustor, fuel nozzles, or exhaust system

Generation - Gas Simple Cycle Task Library (4 of 6)

Activity	Task
Turbine generator system - Generator work	Test, maintain, inspect, or troubleshoot turbine generator
	Rebuild generator
	Inspect, maintain, or operate generator equipment (i.e. PT fuses, turning gear, heaters)
	Calibrate generator
	Calibrate generator instrumentation
	Purge generator of Hydrogen in and out of service
	H2 leak survey, response, or investigation
	Install or remove grounding
	Replace online exciter brush and filter

Generation - Gas Simple Cycle Task Library (5 of 6)

Activity	Task
High voltage electrical system work	Install or remove cable tray and support
	Test electrical equipment
	Test, maintain, inspect, or troubleshoot circuit breaker
	Install or remove equipment or devices (i.e. transformer, cutout, switch, insulator, relay, regulator, capacitor, fuse)
	Test, maintain, inspect, or troubleshoot equipment or devices (i.e. transformer, cutout, switch, insulator, relay, regulator, capacitor, fuse)
	Open or close switch or fuse
	Install or remove wiring
	Install or remove conduit
	Install or remove circuit breaker
	Open or close circuit breaker
	Rack-in or rack-out circuit breaker
Wastewater system work	Install or remove grounding
	Test, inspect, or maintain oil-water separator
	Repair or replace wastewater pumps or coalescing media
	Perform outfall sampling and testing

Generation - Gas Simple Cycle Task Library (6 of 6)

Activity	Task
Fire suppression system work	Install, remove, or replace fire alarm and detection system
	Install, remove, or replace fire suppression system pipe, nozzle, or valve
	Install, remove, or replace smoke or heat detector, pull station or notification device
	Install, remove, or replace sprinkler system
	Install, remove, or replace fire suppression system water supply
	Test or inspect fire alarm and detection system
	Test or inspect fire suppression system pipe, nozzle, and valve
	Test or inspect smoke or heat detector, pull station and notification device
	Test or inspect sprinkler system
	Test or inspect fire suppression system water supply
	Install or remove conduit
	Install or remove wiring
	Terminate and splice wire
	Inspect fire extinguisher
	Test fire doors and dampers
	Test, inspect, or operate fire pumps

Generation - Gas Combined Cycle

Generation - Gas Combined Cycle Task Library (1 of 8)

Activity	Task
Fuel system work	Inspect or maintain fuel storage tanks
	Inspect and calibrate fuel transfer system
	Calibrate fuel system instrumentation
	Repair or replace pipe, flange, T, bypass, or valve - Welding
	Coat pipe or welds
	Test pipe welds - Non-destructive examination
	Test, maintain, inspect, or troubleshoot pipe, flange, T, bypass, or valve
	Pressure test pipe (hydrostatic / pneumatic)
	Leak survey, response, or investigation
	Test, maintain, inspect, or troubleshoot cathodic protection system
	Unload fuel
	Maintain fuel pumps and valves
	Replace gaskets, filters, or regulating valve diaphragms
Water injection system work	Maintain water pumps and valves
	Replace gaskets, filters, or regulating valve diaphragms
	Calibrate water injection system instrumentation
	Operate water treatment system
	Test, maintain, inspect, or troubleshoot water injection system

Generation - Gas Combined Cycle Task Library (2 of 8)

Activity	Task
Continuous emission monitoring system (CEMS) work	Calibrate emissions sensors
	Replace CEMS test gases
	Test CEMS (i.e. stack test, RATA test)
	Repair or replace CEMS umbilical
	Test, maintain, inspect, or troubleshoot water cooling system
	Test, maintain, inspect, or troubleshoot CEMS analyzers
Intake air system work	Replace intake air filters
	Calibrate intake air instrumentation
	Test, maintain, inspect, or troubleshoot intake air system
	Test, maintain, inspect, or troubleshoot evaporative cooling system
	Test, maintain, inspect, or troubleshoot inlet bleed heat system
	Test, clean, or inspect inlet guide vanes
	Inspect, maintain, or operate inlet chiller system
	Install, remove, or replace inlet chiller system components
	Transfer glycol and maintain inlet chiller glycol system
	Test, maintain, inspect, or troubleshoot air filter cleaning system

Generation - Gas Combined Cycle Task Library (3 of 8)

Activity	Task
Turbine generator system - Turbine work	Repair or replace turbine
	Teardown and rebuild turbine
	Test, maintain, inspect, or troubleshoot turbine
	Test, maintain, inspect, or troubleshoot turbine rotor
	Test, maintain, inspect, or troubleshoot turbine fuel and lubrication oil system
	Test, maintain, inspect, or troubleshoot turbine compressor
	Perform offline water wash of turbine compressor
	Calibrate turbine instrumentation
	Balance turbine rotor
	Check turbine alignment
	Calibrate turbine equipment
	Replace gaskets and filters on fuel and lube oil lines
	Align turbine for operations, startup, and shutdown
	Repair or replace combustor, fuel nozzles, or exhaust system
	Test, maintain, inspect, or troubleshoot combustor, fuel nozzles, or exhaust system
Turbine generator system - Steam Turbine work	Inspect, maintain, or operate steam seal system
	Test, maintain, or inspect electric hydraulic control system
	Test or inspect turbine stop and control steam valves

Generation - Gas Combined Cycle Task Library (4 of 8)

Activity	Task
Turbine generator system - Generator work	Test, maintain, inspect, or troubleshoot turbine generator
	Rebuild generator
	Inspect, maintain, or operate generator equipment (PT fuses, turning gear, heaters)
	Calibrate generator
	Calibrate generator instrumentation
	Purge generator of Hydrogen in and out of service
	H2 leak survey, response, or investigation
	Install or remove grounding
	Replace online exciter brush and filter

Generation - Gas Combined Cycle Task Library (5 of 8)

Activity	Task
High voltage electrical equipment work	Install or remove cable tray and support
	Test electrical equipment
	Install or remove equipment or devices (i.e. transformer, cutout, switch, insulator, relay, regulator, capacitor, fuse)
	Test, maintain, inspect, or troubleshoot equipment or devices (i.e. transformer, cutout, switch, insulator, relay, regulator, capacitor, fuse)
	Open or close switch or fuse
	Install or remove wiring
	Install or remove conduit
	Install or remove circuit breaker
	Test, maintain, inspect, or troubleshoot circuit breaker
	Open or close circuit breaker
	Rack-in or rack-out circuit breaker
	Install or remove grounding
Backup power system work	Repair or replace battery system
	Test, maintain, service, inspect, or troubleshoot battery system
	Measure voltage and current on battery bank
	Repair or replace control system
	Test, maintain, inspect, or troubleshoot control system
	Install, remove, or replace control panel and communication system
	Test, maintain, or inspect emergency generator

Generation - Gas Combined Cycle Task Library (6 of 8)

Activity	Task
Fire suppression system work	Install, remove, or replace fire alarm and detection system
	Install, remove, or replace fire suppression system pipe, nozzle, or valve
	Install, remove, or replace smoke or heat detector, pull station or notification device
	Install, remove, or replace sprinkler system
	Install, remove, or replace fire suppression system water supply
	Test or inspect fire alarm and detection system
	Test or inspect fire suppression system pipe, nozzle, and valve
	Test or inspect smoke or heat detector, pull station and notification device
	Test or inspect sprinkler system
	Test or inspect fire suppression system water supply
	Install or remove conduit
	Install or remove wiring
	Terminate and splice wire
	Inspect fire extinguisher
	Test fire doors and dampers
	Test, inspect, or operate fire pumps
Wastewater system work	Test, inspect, or maintain oil-water separator
	Repair or replace wastewater pumps or coalescing media
	Perform outfall sampling and testing

Generation - Gas Combined Cycle Task Library (7 of 8)

Activity	Task
Heat recovery steam generator (HRSG) work	Repair or replace boiler system components
	Test, maintain, inspect, or troubleshoot boiler internally
	Test, maintain, inspect, or troubleshoot boiler externally
	Pressure test boiler (hydrostatic / pneumatic)
	Test, maintain, inspect, or troubleshoot relief devices (i.e. rupture disk, pressure relief valve)
	Test boiler welds - Non-destructive examination
	Test, maintain, inspect, or troubleshoot fittings, manways, flanges, or connections
	Operate boiler system
	Test and maintain boiler water chemistry
	Align HRSG for operations, startup, and shutdown
	Operate boiler feed, condensate, or de-min water pumps
	Operate boiler blowdown system
	Fill or drain HRSG
	Inspect, maintain, or operate chemical feed pumps
	Unload hazardous chemicals (i.e. acid, caustic, ammonia, etc.)
	Inspect, maintain, or operate phosphate injection system
	Inspect, maintain, or operate selective catalytic reduction system
	Inspect, maintain, or operate ammonia storage system and forwarding pumps
	Inspect, maintain, or operate CO2 / VOC catalyst system
	Inspect, maintain, troubleshoot, or operate turbine bypass valves
	Inspect, maintain, troubleshoot, or operate steam attemperators
	Inspect, maintain, troubleshoot, or operate duct burner system

Generation - Gas Combined Cycle Task Library (8 of 8)

Activity	Task
Balance of plant work	Inspect, maintain, troubleshoot, or operate air compressor system
	Inspect, maintain, troubleshoot, or operate closed cooling water system
	Inspect, maintain, troubleshoot, or operate circulating water system
	Inspect, maintain, troubleshoot, or operate cooling tower system
	Inspect, maintain, troubleshoot, or operate cooling tower chemical feed system
	Inspect, maintain, troubleshoot, or operate traveling screens and wash/cleaning system
	Operate, maintain, or troubleshoot auxiliary boiler
Water treatment work	Inspect, maintain, troubleshoot, or operate clarifier and gravity filter system
	Inspect, maintain, troubleshoot, or operate sludge pump and filter press system
	Inspect, maintain, troubleshoot, or operate water treatment system pumps
	Inspect, maintain, troubleshoot, or operate water sampling systems
	Inspect, maintain, troubleshoot, or operate water storage tanks
	Inspect, maintain, troubleshoot, or operate filtration system
	Inspect, maintain, troubleshoot, or operate chemical feed
	Test and calibrate monitoring systems
Steam turbine condenser work	Inspect, maintain, troubleshoot, or operate condenser vacuum system
	Inspect, maintain, troubleshoot, or operate steam jet air ejector
	Inspect, maintain, troubleshoot, or operate chemical feed system
	Inspect, maintain, troubleshoot, or operate water box side of condenser
	Inspect, maintain, or troubleshoot condenser tubes

Generation - Gas Boiler

Generation - Gas Boiler Task Library (1 of 7)

Activity	Task
Fuel system work	Inspect or maintain fuel storage tanks
	Inspect and calibrate fuel transfer system
	Calibrate fuel system instrumentation
	Repair or replace pipe, flange, T, bypass, or valve - Welding
	Coat pipe or welds
	Test pipe welds - Non-destructive examination
	Test, maintain, inspect, or troubleshoot pipe, flange, T, bypass, or valve
	Pressure test pipe (hydrostatic / pneumatic)
	Leak survey, response, or investigation
	Test, maintain, inspect, or troubleshoot cathodic protection system
	Unload fuel
	Maintain fuel pumps and valves
	Replace gaskets, filters, or regulating valve diaphragms
Continuous emission monitoring system (CEMS) work	Calibrate emissions sensors
	Replace CEMS test gases
	Test CEMS (i.e. stack test, RATA test)
	Repair or replace CEMS umbilical
	Test, maintain, inspect, or troubleshoot water cooling system
	Test, maintain, inspect, or troubleshoot CEMS analyzers

Generation - Gas Boiler Task Library (2 of 7)

Activity	Task
Boiler water system work	Repair or replace boiler system components
	Test, maintain, inspect, or troubleshoot boiler internally
	Test, maintain, inspect, or troubleshoot boiler externally
	Pressure test boiler (hydrostatic / pneumatic)
	Test, maintain, inspect, or troubleshoot relief devices (i.e. rupture disk, pressure relief valve)
	Test boiler welds - Non-destructive examination
	Test, maintain, inspect, or troubleshoot fittings, manways, flanges, or connections
	Operate boiler system
	Test and maintain boiler water chemistry
	Align boiler for operations, startup, and shutdown
	Operate boiler feed, condensate, or de-min water pumps
	Operate boiler blowdown system
	Fill or drain boiler
	Inspect, maintain, or operate chemical feed pumps
	Unload hazardous chemicals (i.e. acid, caustic, ammonia, etc.)
	Inspect, maintain, or operate boiler chemical anti-corrosion systems
	Maintain, inspect, troubleshoot soot blower
	Maintain, inspect, or troubleshoot air preheater
	Test, maintain, inspect, or troubleshoot boiler fans (i.e. ID, OFA, recirc, SOFA, LOFA, FD)

Generation - Gas Boiler Task Library (3 of 7)

Activity	Task
Turbine generator system - Turbine work	Repair or replace turbine
	Teardown and rebuild turbine
	Test, maintain, inspect, or troubleshoot turbine
	Test, maintain, inspect, or troubleshoot turbine rotor
	Test, maintain, inspect, or troubleshoot turbine fuel and lubrication oil system
	Test, maintain, inspect, or troubleshoot turbine compressor
	Perform offline water wash of turbine compressor
	Calibrate turbine instrumentation
	Balance turbine rotor
	Check turbine alignment
	Calibrate turbine equipment
	Replace gaskets and filters on fuel and lube oil lines
	Align turbine for operations, startup, and shutdown
	Repair or replace combustor, fuel nozzles, or exhaust system
	Test, maintain, inspect, or troubleshoot combustor, fuel nozzles, or exhaust system
Turbine generator system - Steam Turbine work	Inspect, maintain, or operate steam seal system
	Test, maintain, or inspect electric hydraulic control system
	Test or inspect turbine stop and control steam valves

Generation - Gas Boiler Task Library (4 of 7)

Activity	Task
Turbine generator system - Generator work	Test, maintain, inspect, or troubleshoot turbine generator
	Rebuild generator
	Inspect, maintain, or operate generator equipment (i.e. PT fuses, turning gear, heaters)
	Calibrate generator
	Calibrate generator instrumentation
	Purge generator of Hydrogen in and out of service
	H2 leak survey, response, or investigation
	Install or remove grounding
	Replace online exciter brush and filter
Backup power system work	Repair or replace battery system
	Test, maintain, service, inspect, or troubleshoot battery system
	Measure voltage and current on battery bank
	Repair or replace control system
	Test, maintain, inspect, or troubleshoot control system
	Install, remove, or replace control panel and communication system
	Test, maintain, or inspect emergency generator
Steam turbine condenser work	Inspect, maintain, troubleshoot, or operate condenser vacuum system
	Inspect, maintain, troubleshoot, or operate steam jet air ejector
	Inspect, maintain, troubleshoot, or operate chemical feed system
	Inspect, maintain, troubleshoot, or operate water box side of condenser
	Inspect, maintain, or troubleshoot condenser tubes

Generation - Gas Boiler Task Library (5 of 7)

Activity	Task
High voltage electrical equipment work	Install or remove cable tray and support
	Test electrical equipment
	Install or remove equipment or devices (i.e. transformer, cutout, switch, insulator, relay, regulator, capacitor, fuse)
	Test, maintain, inspect, or troubleshoot equipment or devices (i.e. transformer, cutout, switch, insulator, relay, regulator, capacitor, fuse)
	Open or close switch or fuse
	Install or remove wiring
	Install or remove conduit
	Install or remove circuit breaker
	Test, maintain, inspect, or troubleshoot circuit breaker
	Open or close circuit breaker
	Rack-in or rack-out circuit breaker
Balance of plant work	Install or remove grounding
	Inspect, maintain, troubleshoot, or operate air compressor system
	Inspect, maintain, troubleshoot, or operate closed cooling water system
	Inspect, maintain, troubleshoot, or operate circulating water system
	Inspect, maintain, troubleshoot, or operate cooling tower system
	Inspect, maintain, troubleshoot, or operate cooling tower chemical feed system
	Inspect, maintain, troubleshoot, or operate traveling screens and wash/cleaning system
	Operate, maintain, or troubleshoot auxiliary boiler

Generation - Gas Boiler Task Library (6 of 7)

Activity	Task
Fire suppression system work	Install, remove, or replace fire alarm and detection system
	Install, remove, or replace fire suppression system pipe, nozzle, or valve
	Install, remove, or replace smoke or heat detector, pull station or notification device
	Install, remove, or replace sprinkler system
	Install, remove, or replace fire suppression system water supply
	Test or inspect fire alarm and detection system
	Test or inspect fire suppression system pipe, nozzle, and valve
	Test or inspect smoke or heat detector, pull station and notification device
	Test or inspect sprinkler system
	Test or inspect fire suppression system water supply
	Install or remove conduit
	Install or remove wiring
	Terminate and splice wire
	Inspect fire extinguisher
	Test fire doors and dampers
	Test, inspect, or operate fire pumps

Generation - Gas Boiler Task Library (7 of 7)

Activity	Task
Water treatment work	Inspect, maintain, troubleshoot, or operate clarifier and gravity filter system
	Inspect, maintain, troubleshoot, or operate sludge pump and filter press system
	Inspect, maintain, troubleshoot, or operate water treatment system pumps
	Inspect, maintain, troubleshoot, or operate water sampling systems
	Inspect, maintain, troubleshoot, or operate water storage tanks
	Inspect, maintain, troubleshoot, or operate filtration system
	Inspect, maintain, troubleshoot, or operate chemical feed
	Test and calibrate monitoring systems

Generation - Gas - Reciprocating Internal Combustion Engine (RICE)

Generation - Gas - RICE Task Library (1 of 5)

Activity	Task
Fuel system work	Inspect or maintain fuel storage tanks
	Inspect and calibrate fuel transfer system
	Calibrate fuel system instrumentation
	Repair or replace pipe, flange, T, bypass, or valve - Welding
	Coat pipe or welds
	Test pipe welds - Non-destructive examination
	Test, maintain, inspect, or troubleshoot pipe, flange, T, bypass, or valve
	Pressure test pipe (hydrostatic / pneumatic)
	Leak survey, response, or investigation
	Test, maintain, inspect, or troubleshoot cathodic protection system
	Unload fuel
	Maintain fuel pumps and valves
	Replace gaskets, filters, or regulating valve diaphragms
Continuous emission monitoring system (CEMS) work	Calibrate emissions sensors
	Replace CEMS test gases
	Test CEMS (i.e. stack testing, RATA testing)
	Repair or replace CEMS umbilical
	Test, maintain, inspect, or troubleshoot water cooling system
	Test, maintain, inspect, or troubleshoot CEMS analyzers
	Test, maintain, inspect, or troubleshoot urea system
	Test, maintain, inspect, or troubleshoot selective catalytic reduction

Generation - Gas - RICE Task Library (2 of 5)

Activity	Task
Intake air system work	Replace intake air filters
	Calibrate intake air instrumentation
	Test, maintain, inspect, or troubleshoot intake air system
	Test, maintain, inspect, or troubleshoot evaporative cooling system
	Test, maintain, inspect, or troubleshoot inlet bleed heat system
	Test, clean, or inspect inlet guide vanes
	Inspect, maintain, or operate inlet chiller system
	Install, remove, or replace inlet chiller system components
	Transfer glycol and maintain inlet chiller glycol system
	Test, maintain, inspect, or troubleshoot air filter cleaning system
Backup power system work	Repair or replace battery system
	Test, maintain, service, inspect, or troubleshoot battery system
	Measure voltage and current on battery bank
	Repair or replace control system
	Test, maintain, inspect, or troubleshoot control system
	Install, remove, or replace control panel and communication system
	Test, maintain, or inspect emergency generator

Generation - Gas - RICE Task Library (3 of 5)

Activity	Task
Engine start system work	Test, maintain, inspect, or troubleshoot battery starter motor
	Test, maintain, inspect, or troubleshoot engine start system battery
	Test, maintain, inspect, or troubleshoot air compressor
	Test, maintain, inspect, or troubleshoot air piping
	Test, maintain, inspect, or troubleshoot air starter motor
	Install, remove, or replace battery starter motor
	Install, remove, or replace engine start system battery
	Install, remove, or replace air compressor
	Install, remove, or replace air piping
	Install, remove, or replace air starter motor
Engine generator system work	Repair or replace engine
	Teardown, rebuild, or overhaul engine
	Test, maintain, inspect, or troubleshoot engine (including lubrication oil system)
	Calibrate instrumentation
	Replace gaskets and filters on fuel and lube oil lines
	Test, maintain, inspect, or troubleshoot generator
	Rebuild generator

Generation - Gas - RICE Task Library (4 of 5)

Activity	Task
Fire suppression system work	Install, remove, or replace fire alarm and detection system
	Install, remove, or replace fire suppression system pipe, nozzle, or valve
	Install, remove, or replace smoke or heat detector, pull station or notification device
	Install, remove, or replace sprinkler system
	Install, remove, or replace fire suppression system water supply
	Test or inspect fire alarm and detection system
	Test or inspect fire suppression system pipe, nozzle, and valve
	Test or inspect smoke or heat detector, pull station and notification device
	Test or inspect sprinkler system
	Test or inspect fire suppression system water supply
	Install or remove conduit
	Install or remove wiring
	Terminate and splice wire
	Inspect fire extinguisher
	Test fire doors and dampers
	Test, inspect, or operate fire pumps

Generation - Gas - RICE Task Library (5 of 5)

Activity	Task
High voltage electrical equipment work	Install or remove cable tray and support
	Test electrical equipment
	Install or remove equipment or devices (i.e. transformer, cutout, switch, insulator, relay, regulator, capacitor, fuse)
	Test, maintain, inspect, or troubleshoot equipment or devices (i.e. transformer, cutout, switch, insulator, relay, regulator, capacitor, fuse)
	Open or close switch or fuse
	Install or remove wiring
	Install or remove conduit
	Install or remove circuit breaker
	Test, maintain, inspect, or troubleshoot circuit breaker
	Open or close circuit breaker
	Rack-in or rack-out circuit breaker
	Install or remove grounding
Cooling system work	Test, maintain, inspect, or troubleshoot radiators
	Test, maintain, inspect, or troubleshoot cooling fans
	Test, maintain, inspect, or troubleshoot cooling piping
	Test, maintain, inspect, or troubleshoot pumps
	Install or remove radiators
	Install or remove cooling fans
	Install or remove cooling piping
	Install or remove pumps

Generation - Diesel

Generation - Diesel Task Library (1 of 5)

Activity	Task
Diesel engine work	Test, maintain, inspect, or troubleshoot power packs
	Remove or replace power packs
	Remove or replace exhaust pipe
	Test, maintain, inspect, or troubleshoot injectors
	Remove or replace injectors
	Remove, replace, or inspect engine top deck
	Clean or maintain engine top deck
	Remove or replace filters
	Test, maintain, inspect, or troubleshoot starter
	Remove or replace starter
Accessory rack work	Test, maintain, inspect, or troubleshoot accessories
	Remove or replace water pump
	Remove or replace lube oil pump
	Remove or replace fuel pump
	Remove or replace fuel tank
	Inspect or repair oil filter housing
	Remove or replace thermostat (Amot) valve
	Test, maintain, inspect, or troubleshoot water pump
	Test, maintain, inspect, or troubleshoot lube oil pump
	Test, maintain, inspect, or troubleshoot fuel pump
	Test, maintain, inspect, or troubleshoot fuel tank
	Test, maintain, inspect, or troubleshoot thermostat (Amot) valve

Generation - Diesel Task Library (2 of 5)

Activity	Task
Emissions control system work	Test, maintain, inspect, or troubleshoot analyzers
	Test, maintain, inspect, or troubleshoot pumps
	Remove or replace pumps
	Test, maintain, inspect, or troubleshoot probes
	Test, maintain, inspect, or troubleshoot chillers
	Test, maintain, inspect, or troubleshoot lance
	Test, maintain, inspect, or troubleshoot heated sample line
	Clean, inspect, or replace selective catalytic reduction blocks
	Test, maintain, inspect, or troubleshoot selective catalytic reduction controls
	Maintain, troubleshoot, or update DAHS computer
	Inspect or maintain bulk tanks
	Test, maintain, inspect, or troubleshoot unloading system
	Remove or replace unloading system
	Inspect or maintain piping
	Inspect, maintain, or replace filters
Fueling system work	Inspect or maintain bulk tanks
	Remove or replace external fuel filter
	Inspect or maintain piping
	Test, maintain, inspect, or troubleshoot fuel unloading
	Remove or replace fuel unloading
Generator work	Test, maintain, inspect, or troubleshoot generator

Generation - Diesel Task Library (3 of 5)

Activity	Task
Air system work	Test, maintain, inspect, or troubleshoot air dryer
	Inspect or maintain receiver tank
	Test, maintain, inspect, or troubleshoot compressor
	Inspect or maintain piping
Oil system work	Inspect or maintain transfer pump
	Remove or replace transfer pump
	Inspect or maintain piping
	Inspect or maintain reservoir
Forced air induction system work	Inspect turbo
	Remove or replace turbo
	Inspect supercharger
	Remove or replace supercharger
	Remove or replace oil cooler
	Inspect or maintain piping
	Inspect or maintain pump
	Remove or replace pump
Engine control system work	Test, inspect, maintain, or troubleshoot engine control system
Fire suppression system work	Inspect fire suppression system and fire extinguisher

Generation - Diesel Task Library (4 of 5)

Activity	Task
Fuel system work	Inspect or maintain fuel storage tanks
	Inspect and calibrate fuel transfer system
	Calibrate fuel system instrumentation
	Repair or replace pipe, flange, T, bypass, or valve - Welding
	Coat pipe or welds
	Test pipe welds - Non-destructive examination
	Test, maintain, inspect, or troubleshoot pipe, flange, T, bypass, or valve
	Test, maintain, inspect, or troubleshoot cathodic protection system
	Pressure test pipe (hydrostatic / pneumatic)
	Leak survey, response, or investigation
	Unload fuel
	Maintain fuel pumps and valves
	Replace gaskets, filters, or regulating valve diaphragms

Generation - Diesel Task Library (5 of 5)

Activity	Task
Water injection system work	Maintain water pumps and valves
	Replace gaskets, filters, or regulating valve diaphragms
	Calibrate water injection system instrumentation
	Operate water treatment system
	Test, maintain, inspect, or troubleshoot water injection system
Continuous emission monitoring system (CEMS) work	Calibrate emissions sensors
	Replace CEMS test gases
	Test CEMS (i.e. stack test, RATA test)
	Repair or replace CEMS umbilical
	Test, maintain, inspect, or troubleshoot water cooling system
Intake air system work	Test, maintain, inspect, or troubleshoot CEMS analyzers
	Replace intake air filters
	Calibrate intake air instrumentation
Backup power system work	Test, maintain, inspect, or troubleshoot intake air system
	Repair or replace battery system
	Repair or replace control system
	Test, maintain, service, inspect, or troubleshoot battery system
	Measure voltage and current on battery bank
	Test, maintain, inspect, or troubleshoot control system
	Install, remove, or replace control panel and communication system

Generation - Solid Fuel -
Coal, Biomass, Refuse
Drive Fuel (RDF)

Generation - Solid Fuel Task Library (1 of 7)

Activity	Task
Fuel storage and handling work	Operate, troubleshoot, or maintain fuel handling system
	Repair or replace fuel handling system
	Unload solid fuel from locomotives
	Unload solid fuel from barge
	Pile management
	Reclaim operations
	Respond to storage pile fires
	Operate, troubleshoot or maintain crushers, transfer gates, and mag separators
	Repair or replace crushers, transfer gates, and mag separators
	Operate, troubleshoot or maintain conveyors system
	Repair or replace conveyors system
	Operate, troubleshoot or maintain bunker, trippers, and silos
	Repair or replace bunker, trippers, and silos
	Operate, troubleshoot or maintain yard railroad track and switches
	Repair or replace yard railroad track and switches
	Operate, troubleshoot or maintain coal mills, pulverizers, and blowers
	Repair or replace coal mills, pulverizers, and blowers

Generation - Solid Fuel Task Library (2 of 7)

Activity	Task
Boiler systems work	Operate, troubleshoot or maintain soot blowers
	Repair or replace soot blowers
	Operate, troubleshoot or maintain high pressure / steam piping
	Repair or replace high pressure / steam piping
	Operate, troubleshoot or maintain boiler water systems
	Repair or replace boiler water systems
	Unload hazardous chemicals
	Operate, troubleshoot or maintain boiler water chemical systems
	Repair or replace boiler water chemical systems
	Operate, troubleshoot or maintain boiler water monitoring systems
	Repair or replace boiler water monitoring systems
	Operate, troubleshoot or maintain boiler system fans (FD/ID/OFA)
	Repair or replace boiler system fans (FD/ID/OFA)
	Inspect, maintain, or repair high pressure vessels
	Inspect, maintain, or repair boiler walls, insulation, or shell
	Boiler cleaning and clinker removal

Generation - Solid Fuel Task Library (3 of 7)

Activity	Task
Emissions system work	Unload hazardous chemicals
	Inspect, maintain, or repair ammonia storage tank
	Inspect, maintain, or repair ammonia piping and injection system
	Operate, troubleshoot or maintain selective catalytic reduction and equipment
	Repair or replace selective catalytic reduction and equipment
	Inspect, operate, troubleshoot, or maintain SO2 system
	Repair SO2 system
	Lime system reclaim operations
	Operate, troubleshoot or maintain lime system
	Inspect, maintain, or repair exhaust and stacks
Continuous emission monitoring system (CEMS) work	Calibrate emissions sensors
	Replace CEMS test gases
	Test CEMS (i.e. stack test, RATA test)
	Repair or replace CEMS umbilical
	Test, maintain, inspect, or troubleshoot water cooling system
	Test, maintain, inspect, or troubleshoot CEMS analyzers
	Test, maintain, inspect, or troubleshoot selective catalytic reduction

Generation - Solid Fuel Task Library (4 of 7)

Activity	Task
Ash systems work	Operate, troubleshoot or maintain bottom ash system
	Repair or replace bottom ash system
	Operate, troubleshoot or maintain submerged flight conveyors
	Repair or replace submerged flight conveyors
	Operate, troubleshoot or maintain precipitators, baghouses, and fly ash handling
	Replace baghouse bags
	Operate, troubleshoot or maintain ash handling system
	Repair or replace ash handling system
	Operate, troubleshoot or maintain clinker grinders
	Repair or replace clinker grinders
Turbine generator system - Generator work	Test, maintain, inspect, or troubleshoot turbine generator
	Rebuild generator
	Inspect, maintain, or operate generator equipment (i.e. PT fuses, turning gear, heaters)
	Calibrate generator
	Calibrate generator instrumentation
	Purge generator of Hydrogen in and out of service
	H2 leak survey, response, or investigation
	Install or remove grounding
	Replace online exciter brush and filter

Generation - Solid Fuel Task Library (5 of 7)

Activity	Task
Turbine generator system - Turbine work	Repair or replace turbine
	Teardown and rebuild turbine
	Test, maintain, inspect, or troubleshoot turbine
	Test, maintain, inspect, or troubleshoot turbine rotor
	Test, maintain, inspect, or troubleshoot turbine fuel and lubrication oil system
	Test, maintain, inspect, or troubleshoot turbine compressor
	Perform offline water wash of turbine compressor
	Calibrate turbine instrumentation
	Balance turbine rotor
	Check turbine alignment
	Calibrate turbine equipment
	Replace gaskets and filters on fuel and lube oil lines
	Align turbine for operations, startup, and shutdown
	Repair or replace combustor, fuel nozzles, or exhaust system
	Test, maintain, inspect, or troubleshoot combustor, fuel nozzles, or exhaust system
Wastewater system work	Test, inspect, or maintain oil-water separator
	Repair or replace wastewater pumps or coalescing media
	Perform outfall sampling and testing

Generation - Solid Fuel Task Library (6 of 7)

Activity	Task
High voltage electrical system work	Install or remove cable tray and support
	Test electrical equipment
	Install or remove equipment or devices (i.e. transformer, cutout, switch, insulator, relay, regulator, capacitor, fuse)
	Test, maintain, inspect, or troubleshoot equipment or devices (i.e. transformer, cutout, switch, insulator, relay, regulator, capacitor, fuse)
	Open or close switch or fuse
	Install or remove wiring
	Install or remove conduit
	Install or remove circuit breaker
	Test, maintain, inspect, or troubleshoot circuit breaker
	Open or close circuit breaker
	Rack-in or rack-out circuit breaker
	Install or remove grounding
Backup power system work	Repair or replace battery system
	Test, maintain, service, inspect, or troubleshoot battery system
	Measure voltage and current on battery bank
	Repair or replace control system
	Test, maintain, inspect, or troubleshoot control system
	Install, remove, or replace control panel and communication system
	Test, maintain, or inspect emergency generator

Generation - Solid Fuel Task Library (7 of 7)

Activity	Task
Fire suppression system work	Install, remove, or replace fire alarm and detection system
	Install, remove, or replace fire suppression system pipe, nozzle, or valve
	Install, remove, or replace smoke or heat detector, pull station or notification device
	Install, remove, or replace sprinkler system
	Install, remove, or replace fire suppression system water supply
	Test or inspect fire alarm and detection system
	Test or inspect fire suppression system pipe, nozzle, and valve
	Test or inspect smoke or heat detector, pull station and notification device
	Test or inspect sprinkler system
	Test or inspect fire suppression system water supply
	Install or remove conduit
	Install or remove wiring
	Terminate and splice wire
	Inspect fire extinguisher
	Test fire doors and dampers
	Test, inspect, or operate fire pumps

Generation - Nuclear

Generation - Nuclear Task Library (1 of 7)

Activity	Task
Dry cask storage work	Load or unload fuel
	Weld liner
	Transfer liner to cask
	Transfer cask to storage location
Reactor work	Disassemble or re-assemble reactor
	Remove, repair, or replace reactor closure head
	Remove, repair, or replace upper internal
	Inspect fuel handling
	Load or unload fuel
	Test, maintain, inspect, or troubleshoot instrumentation
	Inspect inservice
	Test, clean, or inspect reactor stud
Reactor coolant pump work	Remove, repair, or replace reactor stud
	Remove, repair, or replace motor
	Test, maintain, inspect, or troubleshoot seal
	Remove, repair, or replace seal
	Test, maintain, inspect, or troubleshoot impeller
	Remove, repair, or replace impeller
	Test, maintain, inspect, or troubleshoot instrumentation
	Inspect or diagnose inservice
	Test pipe welds - Non-destructive examination

Generation - Nuclear Task Library (2 of 7)

Activity	Task
Pressurizer work	Remove, repair, or replace manway
	Test, maintain, inspect, or troubleshoot instrumentation
	Remove, repair, or replace valve
	Test, maintain, inspect, or troubleshoot valve
	Remove, repair, or replace heater
	Test, maintain, inspect, or troubleshoot heater
	Test pipe welds - Non-destructive examination
Steam generator work	Remove, repair, or replace manway or diaphragm
	Install or remove nozzle dam
	Search and retrieval of foreign objects
	Internal test, maintain, inspect, or troubleshoot steam generator
	Sludge lancing activities
	Test, maintain, inspect, or troubleshoot steam generator tube plugging or sleeving
	Test pipe welds - Non-destructive examination
Backup power system work	Repair or replace battery system
	Test, maintain, service, inspect, or troubleshoot battery system
	Measure voltage and current on battery bank
	Repair or replace control system
	Test, maintain, inspect, or troubleshoot control system
	Install, remove, or replace control panel and communication system
	Test, maintain, or inspect emergency generator

Generation - Nuclear Task Library (3 of 7)

Activity	Task
Fuel system - Auxiliary system work	Inspect or maintain fuel storage
	Inspect and calibrate fuel transfer system
	Calibrate fuel system instrumentation
	Repair or replace pipe, flange, T, bypass, or valve - Welding
	Coat pipe or welds
	Test pipe welds - Non-destructive examination
	Test, maintain, inspect, or troubleshoot pipe, flange, T, bypass, or valve
	Pressure test pipe (hydrostatic / pneumatic)
	Leak survey, response, or investigation
	Test, maintain, inspect, or troubleshoot cathodic protection system
	Unload fuel
	Maintain fuel pumps and valves
	Replace gaskets, filters, or regulating valve diaphragms
Wastewater system work	Test, inspect, or maintain oil-water separator
	Repair or replace wastewater pumps or coalescing media
	Perform outfall sampling and testing
Water injection system work	Maintain water pumps, valves, and turbines
	Replace gaskets, filters, or regulating valve diaphragms
	Calibrate water injection system instrumentation
	Operate water treatment system
	Test, maintain, inspect, or troubleshoot water injection system

Generation - Nuclear Task Library (4 of 7)

Activity	Task
Turbine generator system - Turbine work	Repair or replace turbine
	Teardown and rebuild turbine bearings
	Test, maintain, inspect, or troubleshoot turbine
	Test, maintain, inspect, or troubleshoot turbine rotor
	Test, maintain, inspect, or troubleshoot turbine fuel and lubrication oil system
	Test, maintain, inspect, or troubleshoot turbine compressor
	Perform offline water wash of turbine compressor
	Calibrate turbine instrumentation
	Balance turbine rotor
	Check turbine alignment
	Calibrate turbine equipment
	Replace gaskets and filters on fuel and lube oil lines
	Align turbine for operations, startup, and shutdown
	Repair or replace combustor, fuel nozzles, or exhaust system
	Test, maintain, inspect, or troubleshoot combustor, fuel nozzles, or exhaust system

Generation - Nuclear Task Library (5 of 7)

Activity	Task
Turbine generator system - Generator work	Test, maintain, inspect, or troubleshoot turbine generator
	Rebuild generator
	Inspect, maintain, or operate generator equipment (i.e. PT fuses, turning gear, heaters)
	Calibrate generator
	Calibrate generator instrumentation
	Purge generator of Hydrogen in and out of service
	H2 leak survey, response, or investigation
	Install or remove grounding
	Replace online exciter brush and filter

Generation - Nuclear Task Library (6 of 7)

Activity	Task
High voltage electrical system work	Install or remove cable tray and support
	Test electrical equipment
	Install or remove equipment or devices (i.e. transformer, cutout, switch, insulator, relay, regulator, capacitor, fuse)
	Test, maintain, inspect, or troubleshoot equipment or devices (i.e. transformer, cutout, switch, insulator, relay, regulator, capacitor, fuse)
	Open or close switch or fuse
	Install or remove wiring
	Install or remove conduit
	Install or remove circuit breaker
	Test, maintain, inspect, or troubleshoot circuit breaker
	Open or close circuit breaker
	Rack-in or rack-out circuit breaker
	Install or remove grounding

Generation - Nuclear Task Library (7 of 7)

Activity	Task
Fire suppression system work	Install, remove, or replace fire alarm and detection system
	Install, remove, or replace fire suppression system pipe, nozzle, or valve
	Install, remove, or replace smoke or heat detector, pull station or notification device
	Install, remove, or replace sprinkler system
	Install, remove, or replace fire suppression system water supply
	Test or inspect fire alarm and detection system
	Test or inspect fire suppression system pipe, nozzle, and valve
	Test or inspect smoke or heat detector, pull station and notification device
	Test or inspect sprinkler system
	Test or inspect fire suppression system water supply
	Install or remove conduit
	Install or remove wiring
	Terminate and splice wire
	Inspect fire extinguisher
	Test fire doors and dampers
	Test, inspect, or operate fire pumps

Vegetation Management

Vegetation Management Task Library (1 of 2)

Activity	Task
Prep work site	Install or remove work area protection (i.e. traffic control)
	Equipment setup and breakdown
	Operate work area protection (i.e. flaggers)
	Load or unload material or equipment
Aerial work	Remove tree parts from wire under tension - Aerial lift
	Trim vegetation using hand tools - Aerial lift
	Trim vegetation using chainsaw - Aerial lift
	Roping and rigging - Aerial lift
	Tree topping - Aerial lift
	Treat vegetation with herbicide - Aerial lift
	Remove tree parts from wire under tension - Climbing
	Trim vegetation using hand tools - Climbing
	Trim vegetation using chainsaw - Climbing
	Roping and rigging - Climbing
	Tree topping - Climbing

Vegetation Management Task Library (2 of 2)

Activity	Task
Ground work	Tree removal - Felling
	Tree removal - Lodged tree
	Rope and rig vegetation
	Treat vegetation with herbicide - Ground work
	Chip vegetation - Mechanically Fed
	Chip vegetation - Manually Fed
	Remove spring pole
	Remove tree parts from wire under tension
Operate equipment	Winch vegetation
	Operate vegetation management mechanical equipment (i.e. aerial boom saw, feller buncher, mower, log loader, vegetation handling, masticator)
	Grind stump
	Operate remote equipment
	Operate utility terrain vehicle
	Trim vegetation with helicopter
	Chip vegetation - Mechanically Fed
	Chip vegetation - Manually Fed

Fleet Work

Fleet Work Task Library (1 of 5)

Activity	Task
Prep work site	Clear and remove debris, vegetation, or obstructions
	Install or remove work area protection (i.e., traffic control)
	Clean boom components
	Clean vehicle body, chassis, or engine
Vehicle and material handling	Load or unload hazardous material (i.e. receive, stage, or demobilize)
	Load or unload non-hazardous material (i.e. receive, stage, or demobilize)
	Operate forklift
	Hoist engine
	Operate overhead crane
	Rig and lift (i.e. vehicle lift)
Maintain internal combustion engine (ICE) vehicle	Test, maintain, inspect, or troubleshoot engine
	Test, maintain, inspect, or troubleshoot battery 12V
	Install or remove wiring
	Test electrical equipment
	Handle hazardous waste (i.e. coolant, oil, fuel, etc.)
	Adjust, maintain, or repair cooling, exhaust, and fuel systems

Fleet Work Task Library (2 of 5)

Activity	Task
Maintain electric vehicle	Test, maintain, inspect, or troubleshoot battery ESS
	Test, maintain, inspect, or troubleshoot battery 12V
	Discharge capacitors
	Identify high voltage work area
	Isolate high voltage
	Install or remove wiring
	Test electrical equipment
	Lubricate chassis
	Handle hazardous waste (i.e. coolant, oil, fuel, etc.)
Machine and fabrication work	Hot work (i.e. permit work, welding, grinding, use torch)
	Fabricate, alter, or repair equipment or vehicles
HVAC work	Troubleshoot HVAC system (i.e. diagnostics, check system pressures, handle freon)
	Evacuate and charge refrigerant
	Install or remove mechanical compressor
	Install, repair, or replace condenser
	Install, repair, or replace evaporator
	Replace low/high pressure switch
	Install or remove wiring

Fleet Work Task Library (3 of 5)

Activity	Task
Driveline work	Install, repair, or replace transfer case
	Install, repair, or replace driveshaft
	Install, repair, or replace differentials
	Install, repair, or replace transmission
	Install, repair, or replace transmission line
	Install, repair, or replace power takeoff
	Install, repair, or replace bearing
	Install, repair, or replace electric motor
Suspension work	Install, repair, or replace shocks, struts, or springs
	Install, repair, or replace U-bolts
	Install, repair, or replace upper and lower control arms and ball joints
	Install, repair, or replace bump stop (i.e. Timbren rubber springs)
	Install, repair, or replace tracks
	Lubricate chassis
	Install, repair, or replace sway bar or stabilizer
	Install, repair, or replace bearing
	Install, repair, or replace air bags

Fleet Work Task Library (4 of 5)

Activity	Task
Brakes work	Install, remove, or repair hydraulic lines, bleed system, or components (i.e. caliper, master cylinder, wheel cylinder)
	Install, remove, repair, or replace air brake chamber
	Install, remove, repair, or replace air dryer
	Install, remove, repair, or replace air tanks and air lines
	Install, remove, repair, or replace air compressor/governor
	Install, remove, or replace brake linings or hardware
	Install, remove, or replace brake rotors or drums
	Install, repair, or replace bearing
	Install, remove, or replace parking brake cable or hardware
	Install, repair, or replace air brake controller
Trailer work	Install, repair, or replace coupler (i.e. lunette ring, kingpin)
	Install, repair, or replace jack
	Install, repair, or replace auxiliary engine, air filter, spark plugs, wires, oil/filters replacement, chassis lubrication
	Install, repair, or replace bearing
	Fabricate, alter, repair structures
	Install, repair, or replace brakes
	Inspect, maintain, or adjust breakaway brake system

Fleet Work Task Library (5 of 5)

Activity	Task
Aerial device work	Install, repair, replace, or remove bucket or control
	Install, repair, or replace hydraulic cylinder or electric over hydraulic controls and devices
	Install, repair, replace, or diagnose hydraulic line / hose fluid leaks, or system operations
	Install fiber optic signal lines to control boom
	Install, repair, or replace augers or pole guides
	Install, repair, or replace outriggers or interlocks
	Install, repair, or replace winch system (i.e. digger derrick / material lift)
Tires or tracks work	Install, repair, or replace boom components
	Install, repair, or replace tracks
	Install, repair, or replace tire
	Inspect, maintain, or adjust tire pressures
	Balance tires / wheels
	Mount tire to rim

Supply Chain and Warehouse

Supply Chain and Warehouse Task Library (1 of 3)

Activity	Task
Crane Inspections & Operations	Load or unload hazardous material (receive, stage, or demobilize)
	Load or unload non-hazardous material (receive, stage, or demobilize)
	Rig and lift - Non-critical lift
	Rig and lift - Critical lift with plan in place
	Install or remove outrigger pads or cribbing blocks
	Install or remove grounds
	Test, maintain, troubleshoot, or inspect load
	Trial lift (pre-lift)
	Crane assemble or disassemble (climb, platform, deck, trailer, etc.)
Forklift Inspections & Operations	Load or unload hazardous material (receive, stage, or demobilize)
	Load or unload non-hazardous material (receive, stage, or demobilize)
	Rig and lift - Non-critical lift
	Rig and lift - Critical lift with plan in place
	Test, maintain, troubleshoot, or inspect load
	Trial lift (pre-lift)
	Inspect forklift (climb, platform, deck, trailer, etc.)

Supply Chain and Warehouse Task Library (2 of 3)

Activity	Task
Order Picker Inspections & Operations	Load or unload hazardous material (receive, stage, or demobilize)
	Load or unload non-hazardous material (receive, stage, or demobilize)
	Rig and lift - Non-critical lift
	Rig and lift - Critical lift with plan in place
	Test, maintain, troubleshoot, or inspect load
	Trial lift (pre-lift)
	Inspect order picker (climb, platform, deck, trailer, etc.)
	Perform load balance and secure load
	Manually load or unload non-hazardous material
	Manually load or unload hazardous material
	Install or remove forklift attachment
Equipment Attachment Operations	Install or remove order picker attachment
	Install or remove crane attachment (i.e. jib, luffer)
	Operate crane attachment
	Operate forklift attachment
	Operate order picker attachment

Supply Chain and Warehouse Task Library (3 of 3)

Activity	Task
Prep work site	Clear and remove debris, vegetation or obstructions
	Install or remove matting
	Install or remove outrigger pads or cribbing blocks
	Install or remove work area protection (i.e., traffic control, ground barrier)
	Install or remove crane barricades
	Operate work area protection (i.e., flaggers)
Material handling	Load or unload hazardous material (receive, stage, or demobilize)
	Load or unload non-hazardous material (receive, stage, or demobilize)
	Perform load balance and secure load
Material transporting	Operate transport equipment (semi, forklift, moffett)
	Perform load balance and secure load
	Transport light freight (interoffice mail)
Heavy Hauling	Transport oversized loads (specialized equipment greater than 60,000 lbs, large transformers, poles, dozer)
	Load or unload hazardous material (receive, stage, or demobilize)
	Load or unload non-hazardous material (receive, stage, or demobilize)
	Perform load balance and secure load
	Rig and lift - Non-critical lift
	Rig and lift - Critical lift with plan in place
	Measure height of live equipment

Renewable Generation - Solar

Renewable Generation – Solar – Ground Mount Task Library

(1 of 3)

Activity	Task
Site Prep or civil work	Setup and breakdown work modules
	Clear and remove debris, vegetation, trees, or obstructions
	Survey and stake
	Install erosion and sediment (E&S) controls
	Grade (e.g. skid steer, bobcat, etc.)
	Install access roads
	Install, remove, or repair fencing
	Cut/fabricate/modify piles
	Pour concrete pad or install pile
Excavation or trenching	Backfill (tamping, compacting, soil dumping)
	Grade (e.g. skid steer, bobcat, etc.)
	Install or remove protective systems (e.g. shoring, shielding, benching, sloping)
	Install or remove tieback
	Mechanical excavation (e.g. excavator, digger derrick, saw cutting, etc.)
	Non-mechanical excavation (e.g. vacuum truck, hand dig, etc.)
	Restore soft surface (e.g. seed, mulch, lime, sod, etc.)
	Sample soil
	Install or remove conduit
	Install or remove wire

Renewable Generation – Solar – Ground Mount Task Library

(2 of 3)

Activity	Task
Racking work	Survey and stake
	Set pole
	Pile drive
	Install or remove post
	Install or remove racking
	Install or remove tracking system
Module install	Install or remove module or torque tube
	Load or unload material or equipment
	Stage module
	Cut/fabricate/modify torque tube assemblies
	Rig and lift - Non-critical lift
	Rig and lift - Critical lift with plan in place
Connection work	Install module to module connectors
	Connect string to combiner
	Connect combiner to inverter
	Connect DAS equipment
	Install or remove transformer
	Connect transformer
	Install or remove distribution voltage switchgear
	Install or remove transmission voltage switchgear
	Connect distribution voltage switchgear
	Rig and lift - Non-critical lift
	Directional boring
	Connect tracking system

Renewable Generation – Solar – Ground Mount Task Library

(3 of 3)

Activity	Task
Commissioning and interconnection work	Cold test (non-energized)
	Hot test (energized)
	Connect solar system to utility power
	Test, maintain, inspect, or troubleshoot solar system
	Verify site drawing
Preventative maintenance work	Inspect hardware
	Thermal scan
	Clear and remove debris, vegetation, trees, or obstructions
	Current and voltage testing
	Inverter maintenance
	Environmental maintenance
	Site access and patrol station
	Meteorological station tracking system
Troubleshooting or corrective maintenance work	Wire management
	Current and voltage testing
	Install or remove fuse
	Test, maintain, inspect, or troubleshoot connectors
	Test, maintain, inspect or troubleshoot inverters
	Test, maintain, inspect or troubleshoot combiner boxes
	Test, maintain, inspect or troubleshoot tracking system
	Test, maintain, inspect, or troubleshoot modules
	Thermal scan
	Install or remove module
	Test, maintain, inspect or troubleshoot tracker system driveline

Renewable Generation – Solar – Car Port Task Library

(1 of 3)

Activity	Task
Site Prep or civil work	Setup and breakdown work modules
	Clear and remove debris, vegetation, trees, or obstructions
	Survey and stake
	Install erosion and sediment (E&S) controls
	Grade (e.g. skid steer, bobcat, etc.)
	Install, remove, or repair fencing
	Cut/fabricate/modify piles
	Pour concrete pad
Excavation or trenching	Backfill (tamping, compacting, soil dumping)
	Grade (e.g. skid steer, bobcat, etc.)
	Install or remove protective systems (e.g. shoring, shielding, benching, sloping)
	Install or remove tieback
	Mechanical excavation (e.g. excavator, digger derrick, saw cutting, etc.)
	Non-mechanical excavation (e.g. vacuum truck, hand dig, etc.)
	Restore soft surface (e.g. seed, mulch, lime, sod, etc.)
	Sample soil
	Saw cutting
	Directional bore shot

Renewable Generation – Solar – Car Port Task Library

(2 of 3)

Activity	Task
Racking work	Survey and stake
	Predrill foundation post
	Set pole
	Set column
	Set beam
	Build racking
	Install concrete foundation or pier
Module install	Install or remove module
	Load or unload material or equipment
	Stage module
	Cut/fabricate/modify torque tube assemblies
	Rig and lift - Non-critical lift
	Rig and lift - Critical lift with plan in place
Connection work	Install module to module connectors
	Connect string to combiner
	Connect combiner to inverter
	Connect DAS equipment
	Install or remove transformer
	Connect transformer
	Install or remove distribution voltage switchgear
	Connect distribution voltage switchgear
	Rig and lift - Non-critical lift
	Directional boring

Renewable Generation – Solar – Car Port Task Library

(3 of 3)

Activity	Task
Commissioning and interconnection work	Cold test (non-energized)
	Hot test (energized)
	Connect solar system to utility power
	Test, maintain, inspect, or troubleshoot solar system
	Verify site drawing
Preventative maintenance work	Inspect hardware
	Thermal scan
	Clear and remove debris, vegetation, trees, or obstructions
	Current and voltage testing
	Inverter maintenance
	Environmental maintenance
	Site access and patrol station
	Meteorological station tracking system
	Wire management
Troubleshooting or corrective maintenance work	Current and voltage testing
	Install or remove fuse
	Test, maintain, inspect, or troubleshoot connectors
	Test, maintain, inspect, or troubleshoot modules
	Thermal scan
	Test, maintain, inspect or troubleshoot inverters
	Test, maintain, inspect or troubleshoot combiner boxes
	Install or remove module
	Connect replacement

Renewable Generation – Solar – Roof Top Task Library

(1 of 3)

Activity	Task
Site Prep or civil work	Setup and breakdown work modules
	Survey and stake
	Install, remove, or repair fencing
	Pour concrete pad
	Equipment setup and breakdown on roof
	Cut/fabricate/modify piles
	Rig and lift - Non-critical lift
	Rig and lift - Critical lift with plan in place
	Install or remove fall protection system
Excavation or trenching	Backfill (tamping, compacting, soil dumping)
	Install or remove protective systems (e.g. shoring, shielding, benching, sloping)
	Install or remove tieback
	Mechanical excavation (e.g. excavator, digger derrick, saw cutting, etc.)
	Non-mechanical excavation (e.g. vacuum truck, hand dig, etc.)
	Restore soft surface (e.g. seed, mulch, lime, sod, etc.)
	Sample soil
	Install or remove conduit
	Install or remove wire
Racking work	Survey and stake
	Equipment setup and breakdown on roof
	Install or remove ballast
	Install or remove racking

Renewable Generation – Solar – Roof Top Task Library

(2 of 3)

Activity	Task
Module install	Install or remove module or optimizer
	Load or unload material or equipment
	Cut/fabricate/modify torque tube assemblies
Connection work	Install module to module connectors
	Connect string to combiner
	Connect combiner to inverter
	Connect DAS equipment
	Install or remove transformer
	Connect transformer
	Install or remove distribution voltage switchgear
	Connect distribution voltage switchgear
	Rig and lift - Non-critical lift
	Directional boring
Commissioning and interconnection work	Cold test (non-energized)
	Hot test (energized)
	Connect solar system to utility power
	Test, maintain, inspect, or troubleshoot solar system
	Verify site drawing

Renewable Generation – Solar – Roof Top Task Library

(3 of 3)

Activity	Task
Preventative maintenance work	Inspect hardware
	Thermal scan
	Clear and remove debris, vegetation, trees, or obstructions
	Inverter maintenance
	Environmental maintenance
	Site access and patrol station
	Meteorological station tracking system
	Wire management
	Current and voltage testing
Troubleshooting or corrective maintenance work	Current and voltage testing
	Install or remove fuse
	Test, maintain, inspect, or troubleshoot connectors
	Test, maintain, inspect or troubleshoot inverters
	Test, maintain, inspect or troubleshoot combiner boxes
	Test, maintain, inspect, or troubleshoot modules
	Thermal scan
	Install or remove module
	Connect replacement

Renewable Generation – Solar – Floating/Over water

Task Library (1 of 3)

Activity	Task
Site Prep or civil work	Setup and breakdown work modules
	Survey and stake
	Install erosion and sediment (E&S) controls
	Grade (e.g. skid steer, bobcat, etc.)
	Install access roads
	Install, remove, or repair fencing
	Cut/fabricate/modify piles
	Pour concrete pad
Excavation or trenching	Backfill (tamping, compacting, soil dumping)
	Grade (e.g. skid steer, bobcat, etc.)
	Install or remove protective systems (e.g. shoring, shielding, benching, sloping)
	Install or remove tieback
	Mechanical excavation (e.g. excavator, digger derrick, saw cutting, etc.)
	Non-mechanical excavation (e.g. vacuum truck, hand dig, etc.)
	Restore soft surface (e.g. seed, mulch, lime, sod, etc.)
	Sample soil
Anchor work	Install or remove anchor
	Drive anchor
	Dive operations for anchor inspections
	Survey and stake
Racking work	Assemble pontoon or racking
	Install or remove racking

Renewable Generation – Solar – Floating/Over water

Task Library (2 of 3)

Activity	Task
Module install	Install or remove module
	Cut/fabricate/modify torque tube assemblies
Connection work	Install module to module connectors
	Connect string to combiner
	Connect combiner to inverter
	Connect DAS equipment
	Install or remove transformer
	Connect transformer
	Install or remove distribution voltage switchgear
	Connect distribution voltage switchgear
	Directional boring
Commissioning and interconnection work	Cold test (non-energized)
	Hot test (energized)
	Connect solar system to utility power
	Test, maintain, inspect, or troubleshoot solar system
	Verify site drawing
Preventative maintenance work	Inspect hardware
	Thermal scan
	Clear and remove debris, vegetation, trees, or obstructions
	Dive operations for PM
	Water management
	Current and voltage testing

Renewable Generation – Solar – Floating/Over water

Task Library (3 of 3)

Activity	Task
Troubleshooting or corrective maintenance work	Current and voltage testing
	Install or remove fuse
	Test, maintain, inspect, or troubleshoot connectors
	Test, maintain, inspect, or troubleshoot modules
	Thermal scan
	Test, maintain, inspect or troubleshoot inverters
	Test, maintain, inspect or troubleshoot combiner boxes
	Install or remove module
	Connect replacement

Renewable Generation - Wind

Renewable Generation – Wind Task Library (1 of 4)

Activity	Task
Prep work site	Clear and remove debris, vegetation, trees, or obstructions
	Install or remove fall protection system
	Install or remove live line insulation, cover up, or rubber goods
	Locate or mark-out
	Load or unload material or equipment
Material handling / hoist work	Load or unload material or equipment
	Rig and lift - Non-critical lift
	Rig and lift - Critical lift with plan in place
Hub and blade work	Enter or work in confined or enclosed space
	Install or remove cable brackets or components
	Install or remove vault, manhole, or pad
	Maintain or inspect blades
	Maintain or inspect hub
Excavate	Backfill (tamping, compacting, soil dumping)
	Grade (e.g. skid steer, bobcat, etc.)
	Install or remove protective systems (e.g. shoring, shielding, benching, sloping)
	Install or remove tieback
	Mechanical excavation (e.g. excavator, digger derrick, saw cutting, etc.)
	Non-mechanical excavation (e.g. vacuum truck, hand dig, etc.)
	Restore soft surface (e.g. seed, mulch, lime, sod, etc.)
	Sample soil

Renewable Generation – Wind Task Library (2 of 4)

Activity	Task
Wire work	Fault locate on wire
	Install or remove live line insulation, cover up, or rubber goods
	Install, remove, or transfer wire
	Terminate and splice wire
	Test, maintain, inspect, or troubleshoot wire
	Install or remove grounding
Turbine generator system - Turbine work	Repair or replace turbine
	Teardown and rebuild turbine
	Test, maintain, inspect, or troubleshoot turbine
	Test, maintain, inspect, or troubleshoot turbine rotor
	Test, maintain, inspect, or troubleshoot lubrication oil system
	Test, maintain, inspect, or troubleshoot turbine compressor
	Calibrate turbine instrumentation
	Balance turbine rotor
	Check turbine alignment
	Calibrate turbine equipment
	Replace gaskets and filters on lube oil lines
	Align turbine for operations, startup, and shutdown
	Refill nitrogen tanks
	Test fire suppression equipment
	Test, maintain, inspect, or troubleshoot nitrogen tanks

Renewable Generation – Wind Task Library (3 of 4)

Activity	Task
Turbine generator system - Generator work	Test, maintain, inspect, or troubleshoot turbine generator
	Rebuild generator
	Inspect, maintain, or operate generator equipment (i.e. fuses, turning gear, heaters,)
	Calibrate generator
	Calibrate generator instrumentation
	Install or remove grounding
	Replace brush and filter
High voltage electrical system work	Install or remove cable tray and support
	Test electrical equipment
	Test, maintain, inspect, or troubleshoot circuit breaker
	Install or remove equipment or devices (i.e. transformer, switch, insulator, relay, regulator, capacitor, fuse)
	Test, maintain, inspect, or troubleshoot equipment or devices (i.e. transformer, switch, insulator, relay, regulator, capacitor, fuse)
	Open or close switch or fuse
	Install or remove wiring
	Install or remove conduit
	Install or remove circuit breaker
	Open or close circuit breaker
	Rack-in or rack-out circuit breaker
	Install or remove grounding

Renewable Generation – Wind Task Library (4 of 4)

Activity	Task
Foundation and civil work	Install, remove, or repair pad or foundation
	Install, remove, or repair spill containment
	Install or remove conduit, duct bank, or cable trenches
	Fabricate, alter, assemble, or remove structure
	Install, remove, or repair fencing

Renewable Generation - Hydroelectric

Renewable Generation – Hydroelectric Task Library (1 of 7)

Activity	Task
Cooling water system work	Maintain water pumps and valves
	Replace gaskets, filters, or regulating valve diaphragms
	Calibrate water system instrumentation
	Operate water treatment system
	Test, maintain, inspect, or troubleshoot water system
Backup power system work	Repair or replace battery system
	Repair or replace control system
	Test, maintain, service, inspect, or troubleshoot battery system
	Measure voltage and current on battery bank
	Test, maintain, inspect, or troubleshoot control system
	Install, remove, or replace control panel and communication system
	Test, maintain, or inspect emergency generator
	Test backup power equipment from secondary source

Renewable Generation – Hydroelectric Task Library (2 of 7)

Activity	Task
Turbine generator system - Turbine work	Replace turbine
	Teardown and rebuild turbine
	Test, maintain, inspect, or troubleshoot turbine
	Test, maintain, inspect, or troubleshoot turbine rotor
	Test, maintain, inspect, or troubleshoot lubrication oil system
	Test, maintain, inspect, or troubleshoot turbine compressor
	Calibrate turbine instrumentation
	Balance turbine rotor
	Check turbine alignment
	Calibrate turbine equipment
	Replace gaskets and filters on lube oil lines
	Align turbine for operations, startup, and shutdown
Turbine generator system - Generator work	Test, maintain, inspect, or troubleshoot turbine generator
	Rebuild generator
	Inspect, maintain, or operate generator equipment (i.e. PT fuses, turning gear, heaters)
	Calibrate generator
	Calibrate generator instrumentation
	Install or remove grounding
	Replace online exciter brush and filter

Renewable Generation – Hydroelectric Task Library (3 of 7)

Activity	Task
High voltage electrical system work	Install or remove cable tray and support
	Test electrical equipment
	Test, maintain, inspect, or troubleshoot circuit breaker
	Install or remove equipment or devices (i.e. transformer, cutout, switch, insulator, relay, regulator, capacitor, fuse)
	Test, maintain, inspect, or troubleshoot equipment or devices (i.e. transformer, cutout, switch, insulator, relay, regulator, capacitor, fuse)
	Open or close switch or fuse
	Install or remove wiring
	Install or remove conduit
	Install or remove circuit breaker
	Open or close circuit breaker
	Rack-in or rack-out circuit breaker
	Install or remove grounding
Wastewater system work	Test, inspect, or maintain oil-water separator
	Repair or replace wastewater pumps or coalescing media
	Perform outfall sampling and testing

Renewable Generation – Hydroelectric Task Library (4 of 7)

Activity	Task
Fire suppression system work	Install, remove, or replace fire alarm and detection system
	Install, remove, or replace fire suppression system pipe, nozzle, or valve
	Install, remove, or replace smoke or heat detector, pull station or notification device
	Install, remove, or replace sprinkler system
	Install, remove, or replace fire suppression system water supply
	Test or inspect fire alarm and detection system
	Test or inspect fire suppression system pipe, nozzle, and valve
	Test or inspect smoke or heat detector, pull station and notification device
	Test or inspect sprinkler system
	Test or inspect fire suppression system water supply
	Install or remove conduit
	Install or remove wiring
	Terminate and splice wire
	Inspect fire extinguisher
	Test fire doors and dampers
	Test, inspect, or operate fire pumps
Cooling water pump work	Remove, repair, or replace motor
	Test, maintain, inspect, or troubleshoot seal
	Remove, repair, or replace seal
	Test, maintain, inspect, or troubleshoot impeller
	Remove, repair, or replace impeller
	Test, maintain, inspect, or troubleshoot instrumentation
	Inspect or diagnose inservice
	Test pipe welds - Non-destructive examination

Renewable Generation – Hydroelectric Task Library (5 of 7)

Activity	Task
Prep Work Site	Assemble or demo scaffold
	Clear and remove debris, vegetation, trees, or obstructions
	Install or remove fall protection system
	Install or remove matting
	Install or remove work area protection (i.e. traffic control)
	Operate work area protection (i.e. flaggers)
	Locate or mark-out
	Survey and stake
	Load or unload material or equipment
	Install or remove road plates
Material Handling	Load or unload material or equipment
	Rig and lift - Non-critical lift
	Rig and lift - Critical lift with plan in place
	Rig and lift - Non-critical lift on barge
	Rig and lift - Critical lift with plan in place on barge

Renewable Generation – Hydroelectric Task Library (6 of 7)

Activity	Task
Excavate	Backfill (tamping, compacting, soil dumping)
	Grade (e.g. skid steer, bobcat, etc.)
	Install or remove protective systems (e.g. shoring, shielding, benching, sloping)
	Install or remove tieback
	Mechanical excavation (e.g. excavator, digger derrick, etc.)
	Non-mechanical excavation (e.g. vacuum truck, hand dig, etc.)
	Restore soft surface (e.g. seed, mulch, lime, sod, etc.)
	Install, remove, or repair paving or asphalt
	Sample soil
	Monitor atmospheric conditions
	Load or unload material or equipment
Intake debris clearing	Clearing debris with trash rack, stationary crane, or grapple crane
	Clearing debris with mobile crane
Waterway clearing	Clearing debris or dredging with boats / mobile equipment
	Clearing debris with debris boom
	Dredging operations with divers
Dive inspection work	Diving inspection
	Inspection with ROV (remote operated vehicle)
Gate and valve work	Test, maintain, inspect, or troubleshoot gate or valve
	Install, remove, replace, or rehabilitate gate or valve
	Operate gate or valve
	Install or remove coffer dam
	Install or remove stop log

Renewable Generation – Hydroelectric Task Library (7 of 7)

Activity	Task
Penstock work	Install, remove, or repair penstock
	Test, maintain, inspect, or troubleshoot penstock
Tunnel work	Install, remove, or repair tunnel
	Maintain, inspect, or troubleshoot tunnel or instrumentation
Dam and diversion work	Maintain, inspect, or troubleshoot stilling well or pizometer well
	Flush gallery or stilling well
	Clean or flush pizometer
	Maintain, repair, or inspect dam

Renewable Generation - Battery

Renewable Generation – Battery Task Library (1 of 6)

Activity	Task
Site Prep or civil work	Construct foundation
	Collect geotechnical samples
	Clear and remove debris, vegetation, trees, or obstructions
	Survey and stake
	Install erosion and sediment (E&S) controls
	Grade (e.g. skid steer, bobcat, etc.)
	Install access roads
	Install, remove, or repair fencing
	Pour concrete pad
Excavation or trenching	Backfill (tamping, compacting, soil dumping)
	Grade (e.g. skid steer, bobcat, etc.)
	Install or remove protective systems (e.g. shoring, shielding, benching, sloping)
	Install or remove tieback
	Mechanical excavation (e.g. excavator, digger derrick, saw cutting, etc.)
	Non-mechanical excavation (e.g. vacuum truck, hand dig, etc.)
	Restore soft surface (e.g. seed, mulch, lime, sod, etc.)
	Sample soil
	Install or remove conduit
	Install or remove wire

Renewable Generation – Battery Task Library (2 of 6)

Activity	Task
Battery storage system work	Install or remove module
	Install or remove rack or battery container
	Load or unload material or equipment
	Stage module
	Stage rack or battery container
	Rig and lift - Non-critical lift
	Rig and lift - Critical lift with plan in place
	Setup and breakdown work modules
	Setup and breakdown rack or battery container
	Install or remove control system (i.e. SCADA, control house, HMI, battery monitoring, fire control panel, etc.)
	Install or replace coolant pump
	Install or replace coolant fan
	Install or replace cooling piping/fittings
Indoor storage system work	Install or remove module
	Install or remove rack or battery container
	Load or unload material or equipment
	Stage module
	Stage rack or battery container
	Rig and lift - Non-critical lift
	Rig and lift - Critical lift with plan in place
	Setup and breakdown work modules
	Setup and breakdown rack or battery container
	Install or remove control system (i.e. SCADA, control house, HMI, battery monitoring, fire control panel, etc.)

Renewable Generation – Battery Task Library (3 of 6)

Activity	Task
Outdoor storage system work	Install or remove module
	Install or remove rack or battery container
	Load or unload material or equipment
	Stage module
	Stage rack or battery container
	Rig and lift - Non-critical lift
	Rig and lift - Critical lift with plan in place
	Setup and breakdown work modules
	Setup and breakdown rack or battery container
	Install or remove control system (i.e. SCADA, control house, HMI, battery monitoring, fire control panel, etc.)
Connection work	Install module to module connectors
	Connect rack to rack connectors
	Connect system container/unit to inverter
	Connect DAS equipment
	Install or remove transformer
	Connect transformer
	Install or remove distribution voltage switchgear
	Install or remove transmission voltage switchgear
	Connect distribution voltage switchgear
	Rig and lift - Non-critical lift
	Directional boring

Renewable Generation – Battery Task Library (4 of 6)

Activity	Task
Commissioning and interconnection work	Cold test (non-energized)
	Hot test (energized)
	Connect battery storage system to utility power
	Test, maintain, inspect, or troubleshoot battery storage system
	Verify site drawing
Preventative maintenance work	Inspect hardware
	Thermal scan
	Clear and remove debris, vegetation, trees, or obstructions
	Current and voltage testing
	Inverter maintenance
	Environmental maintenance
	Site access and patrol station
	Wire management
	Battery maintenance or cell management
	Drain or refill coolant system
	PM coolant systems (filters, intakes, etc.)
	Testing voltages on coolant system controls

Renewable Generation – Battery Task Library (5 of 6)

Activity	Task
Troubleshooting or corrective maintenance work	Current and voltage testing
	Install or remove fuse
	Test, maintain, inspect, or troubleshoot connectors
	Test, maintain, inspect or troubleshoot inverters
	Test, maintain, inspect or troubleshoot string or rack controllers
	Test, maintain, inspect, or troubleshoot modules
	Thermal scan
	Install or remove module
	Test, maintain, inspect or troubleshoot hydrogen detection system
	Install or replace coolant pump
	Install or replace coolant fan
	Install or replace cooling piping/fittings

Renewable Generation – Battery Task Library (6 of 6)

Activity	Task
Fire suppression system work	Package or recycle battery storage module
	Install, remove, or replace fire alarm and detection system
	Install, remove, or replace fire suppression system pipe, nozzle, or valve
	Install, remove, or replace smoke or heat detector, pull station or notification device
	Install, remove, or replace sprinkler system
	Install, remove, or replace fire suppression system water supply
	Test or inspect fire alarm and detection system
	Test or inspect fire suppression system pipe, nozzle, and valve
	Test or inspect smoke or heat detector, pull station and notification device
	Test or inspect sprinkler system
	Test or inspect fire suppression system water supply
	Install or remove conduit
	Install or remove wiring
	Terminate and splice wire
	Inspect fire extinguisher
	Test fire doors and dampers
	Test, inspect, or operate fire pumps
	Install, remove, or replace dry agent fire suppression system
	Test or inspect dry agent fire suppression system
HVAC system work	Install or remove HVAC system
	Test, maintain, inspect or troubleshoot HVAC system
	Install or remove back up power supply
	Test, maintain, inspect or troubleshoot back up power supply